

YIMO 2026: Finals Round

Window A

Division 1

Format: 4 problems: 2 computational and 2 proof.

Time: 75 minutes. Computational answers are positive integers. Proofs must be complete and rigorous.

1. Let a, b, c be positive real numbers satisfying

$$a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \quad \text{and} \quad a^2 + b^2 + c^2 = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 6.$$

The value of $a + b + c$ can be expressed in the form $p + \sqrt{q}$ for positive integers p and q . Find $p + q$.

2. In triangle ABC , $AB = 9$, $BC = 10$, and $CA = 11$. The incircle of ABC touches side BC at point X . Line AX meets the incircle again at a second point Y (distinct from X). The length XY^2 can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m + n$.
3. Determine, with proof, all pairs of positive integers (a, b) such that both $a^2 + b$ and $b^2 + a$ are perfect squares.
4. The cells of an $n \times n$ array contain real numbers (not necessarily distinct), and the sum of all entries is positive. In one move, you may multiply every entry of a single row by -1 , or every entry of a single column by -1 . Prove that it is always possible, after finitely many moves, to reach an arrangement in which the sum of the entries in every row and every column is nonnegative.

YIMO 2026: Finals Round

Window A

Division 2

Format: 4 problems: 3 computational and 1 proof.

Time: 60 minutes. Computational answers are positive integers. The proof must be complete and rigorous.

1. Find the sum of all real solutions x to the equation

$$x^4 + (x - 2)^4 = 82.$$

2. Find the number of positive integers n with $1 \leq n \leq 1000$ such that $n^2 + n + 1$ is divisible by 7.
3. In acute triangle ABC , $AB = 5$, $AC = 7$, and the altitude from A to side BC has length 4. The length BC can be expressed in the form $\sqrt{m} + \sqrt{n}$ where m and n are positive integers with $m \leq n$. Find $m + n$.
4. Eight people stand in a circle. Each person silently chooses a positive integer and writes it on a card. Then each person announces the sum of the two numbers written by their two immediate neighbors. The announced numbers, going around the circle in order, are exactly 1, 2, 3, 4, 5, 6, 7, 8.

Determine, with proof, whether such a configuration is possible.

YIMO 2026: Finals Round

Window B

Division 1

Format: 4 problems: 2 computational and 2 proof.

Time: 75 minutes. Computational answers are positive integers. Proofs must be complete and rigorous.

1. Let a, b, c be positive real numbers. Prove that

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{a+b+c}{2}.$$

Determine, with proof, all triples (a, b, c) for which equality holds.

2. In triangle ABC , $\angle BAC = 60^\circ$, $AB = 5$, and $AC = 8$. The angle bisector from A meets segment BC at D , and the perpendicular from D to line AB meets AB at E . The length DE^2 can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m + n$.

3. Let p be an odd prime. Prove that

$$\binom{2p}{p} \equiv 2 \pmod{p^2}.$$

4. A prototype Artemis lunar landing platform has five landing legs: four legs L_1, L_2, L_3, L_4 positioned at the vertices of a square (labeled in order around the square), and a fifth leg L_5 positioned at the center of the square. Each leg initially has length 10.

For how many ordered 5-tuples $(k_1, k_2, k_3, k_4, k_5)$ of nonnegative integers can engineers shorten leg L_i by k_i units for each $i = 1, 2, 3, 4, 5$ and still have the platform rest stably on a flat lunar surface?

(The platform is stable if all five footpads can simultaneously touch the ground. A final leg length of 0 is permitted.)

YIMO 2026: Finals Round

Window B

Division 2

Format: 4 problems: 3 computational and 1 proof.

Time: 60 minutes. Computational answers are positive integers. The proof must be complete and rigorous.

1. The polynomial $x^4 - 4x^3 + 6x^2 - 4x - 2024$ has four roots r_1, r_2, r_3, r_4 (counted with multiplicity, possibly complex). Find the value of $r_1^4 + r_2^4 + r_3^4 + r_4^4$.
2. Find the largest positive integer n less than 2026 such that $n^3 - n$ is divisible by 504.
3. In triangle ABC , $AB = 14$, $BC = 13$, and $CA = 15$. Let O be the circumcenter and I be the incenter of triangle ABC . The value of OI^2 can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m + n$.
4. A 10×10 board is to be tiled completely by 25 non-overlapping 1×4 tiles, where each tile may be placed either horizontally or vertically. Prove that no such tiling exists.